

Weekly Self-Study Tasks to Complete

Introduction to NoSQL Databases for Leaders

This document covers all of Week 4 of Phase 3 in the 4-hour Leaders model: 2 hours of instructor-led session, 1 hour of self-study video, and 1 hour of hands-on lab. The Core Resources table below fits inside the 1-hour self-study media budget. The Practice Activities section points to the Evergreen Insurance Quote Data lab, where you will store and query the quotes in MongoDB. Additional Content at the end is optional and sits outside the 4-hour budget.

Core Resources (~60 min)

Watch the specific resources called out below — not an entire course. Keep mongosh open as you watch so you can run each command the videos show.

Topic	Material Description	Est. Time	Link / Reference
NoSQL Big Picture	NoSQL: The Big Picture (Andrew Brust, Pluralsight, updated Oct 12, 2024; 1h 10m course). Watch “What is NoSQL?” (~20 min) and “Is NoSQL Right for Your Organization?” (~13 min) — an executive briefing on what NoSQL is and when it fits versus relational. Concept-only.	~30 min	pluralsight.com — NoSQL: The Big Picture
MongoDB Basics	MongoDB Manual “Introduction” and “CRUD Operations” (official docs) — read how documents, collections, and basic find/insert work.	~15 min	mongodb.com/docs — Manual
CRUD & Aggregation	MongoDB University “Introduction to MongoDB” (free) — the official course units on find, update, and a first aggregation.	~15 min	learn.mongodb.com — Introduction to MongoDB

Practice Activities (1 hour)

Complete this week's hands-on lab: "Evergreen Insurance Quote Data — Storing and Querying in MongoDB." You will load the quotes into MongoDB and query them with mongosh — filtering, projecting, updating, aggregating, and indexing. The whole exercise takes about one hour.

In the lab you will:

- Open the Labfiles/evergreen-quote-mongodb-data folder and load the data with mongosh evergreen seed.js.
- Open the mongosh shell and find all quotes, then filter to only auto quotes.
- Use a comparison operator (\$gte) to find higher-premium quotes, and a projection to pick fields.
- Update a quote with \$set and add a tag with \$push, matching on the unique _id.
- Run an aggregation that counts and averages the monthly premium by coverage type.
- Create an index on type and use explain to see the query switch from COLLSCAN to IXSCAN.

The lab instructions (with "Why this matters" callouts throughout) are in the lab environment. Budget one hour and work at your own pace.

Additional Content (Optional)

The following resources are optional and recommended for students who want to go deeper beyond the core curriculum. They are outside the 4-hour weekly budget. Next week connects this database to the Express API, so focus this week on documents, queries, and indexes.

Topic	Material Description	Link / Reference
MongoDB	MongoDB Essential Training (LinkedIn Learning) — a complete course covering documents, CRUD, indexes, and aggregation in depth (Video Course).	linkedin.com/learning – MongoDB Essential Training

Topic	Material Description	Link / Reference
Free Course	MongoDB University — free, hands-on courses and labs from the makers of MongoDB, with a built-in sandbox (Interactive).	learn.mongodb.com
Reference	The MongoDB Manual — the authoritative reference for CRUD, query operators, aggregation, and indexes (Reference).	mongodb.com/docs/manual
DynamoDB	Amazon DynamoDB Developer Guide — core concepts (tables, items, keys) for the other major NoSQL store mentioned this week (Reference).	docs.aws.amazon.com — DynamoDB
Data Modeling	MongoDB Data Modeling Introduction — the official guide to embedding vs referencing and designing for your access patterns (Tutorial).	mongodb.com/docs — Data Modeling